

On Prime Counting Functions Using Odd K -almost Primes: An Exposition

by

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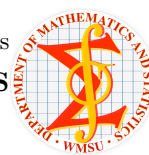
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APPROVAL SHEET

This is to certify that this undergraduate special project entitled “**On Prime Counting Functions Using Odd K-almost Primes: An Exposition**”, prepared and submitted by **Bunnie Ruth R. Aquino** to fulfill part of the requirements for the degree of **Bachelor of Science in Mathematics**, was successfully defended and approved on April 10, 2025 .

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Abstract

On Prime Counting Functions Using Odd K-almost Primes: An Exposition

by Bunnie Ruth R. Aquino

This study discusses in detail the iterative process for determining the number of primes less than or equal to n , the number of 2-almost primes less than or equal to n , and the number of odd 2-almost primes less than or equal to n . Furthermore, it derives formulas for calculating the number of 2-almost primes, odd 2-almost primes, 3-almost primes, and odd 3-almost primes less than or equal to n .

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capable of overcoming anything. I am proud of the person I've become, and I know that **Banban**, the child version of me who once dreamed of this moment must be so proud of how far I've come. This is just the beginning of what I can achieve.

This chapter may be coming to an end, but the lessons I've learned, the memories I've made, and the relationships I've built along the way will stay with me forever. Every challenge, triumph, and moment of growth has shaped me into who I am today. As this journey concludes, I step into the next with a heart overflowing with gratitude and a mind prepared to embrace whatever lies ahead. The end of this chapter is just the beginning of a new adventure filled with endless possibilities.

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List of Notations

Notation	Description
$\mathbb{R}$	Set of Real Numbers
$\mathbb{N}$	Set of Natural Numbers
$\mathbb{Z}$	Set of Integers
$<$	Less than
$>$	Greater than
$\leq$	Less than or equal
$\geq$	Greater than or equal
$\sum$	Summation of
$ A $	Cardinality of A
$\lfloor A \rfloor$	Floor function of A
p_i	i th prime number
$\pi(n)$	Number of primes $\leq n$
$\pi^k(n)$	Number of k -almost primes $\leq n$
$\Pi(n)$	Number of odd primes $\leq n$
$\Pi_m(n)$	$m \leq$ number of odd primes $\leq n$.
$\Pi^k(n)$	Number of odd k -almost primes $\leq n$
$\Pi_m^k(n)$	$m \leq$ number of odd k -primes $\leq n$.

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Chapter 1

Introduction

Prime numbers have long captivated mathematicians and scholars due to their fundamental role in number theory. A prime number is defined as a natural number greater than one that has no positive divisors other than one and itself. Primes are the building blocks of the integers. The distribution of primes among the natural numbers has been a subject of extensive study, leading to significant discoveries that have shaped modern mathematics.

At the heart of Number Theory lies the prime-counting function, denoted as $\pi(n)$, which counts the number of prime numbers less than or equal to a given integer n . The significance of $\pi(n)$ extends beyond mere enumeration; it provides insights into the distribution of primes. The Prime Number Theorem (PNT) not only highlights the asymptotic behavior of primes but also serves as a cornerstone for further investigations about prime distribution.

Numerous approaches have been developed in an attempt to provide precise formulas for $\pi(n)$. Among these are Legendre's formula, Lehmer's formula, Meissel's formula, and Mapes' method. Legendre's formula, introduced in the late 18th century, uses inclusion-exclusion but is not practical for large n due to its computational complexity. Meissel's formula builds on Legendre's method, significantly reducing the number of divisions needed and allowing for efficient calculations up to one billion. Further refinement comes with Lehmer's formula, which employs advanced computational techniques to enable efficient calculations with reduced overhead. Finally, Mapes' method integrates modern computational capabilities, optimizing memory and processing speed to compute $\pi(n)$ rapidly for very large values of n .

Despite these advancements, no single universally accepted formula has emerged that yields precise result in finding the number of primes less than or equal to n .

In recent years, the study of k -almost primes has emerged as a vital area of research within number theory. A k -almost prime is a natural number that has exactly k prime factors, counted with multiplicity. This classification allows for a comprehensive understanding of integer factorization and its implications for prime counting. Conway et al. define "1-almost prime" numbers as equivalent to prime numbers, "2-almost prime" numbers correspond to semiprimes (or biprimes), "3-almost primes" refer to triprimes, and "4-almost primes" are known as quadprimes.

Among k -almost primes, odd k -almost primes warrant particular attention due to their unique properties and potential applications in calculating $\pi(n)$. Unlike even k -almost primes, which often include factors of 2, odd k -almost primes provide a distinct perspective on prime factorization and distribution. By focusing on these numbers, researchers can develop more refined methods for estimating and understanding the behavior of the prime-counting function.

Despite existing literature on both prime counting functions and k -almost primes, there remains a significant gap in understanding how odd k -almost primes can enhance calculations of $\pi(n)$. The main objective of the study is to determine the number of k -almost primes and odd k -almost primes less than or equal to n , for $k = 2$ and 3 . Specifically:

1. Provide an iteration on finding the number of primes less than or equal to n .
2. Provide an iteration on finding the number of 2-almost primes less than or equal to n .
3. Provide an iteration on finding the number of odd 2-almost primes less than or equal to n .

4. Determine the number of 2-almost primes and 3-almost less than or equal to n
5. Determine the number of odd 2-almost and odd 3-almost primes less than or equal to n
6. Provide a detailed proof on the number of 2-almost primes and 3-almost less than or equal to n .
7. Provide a detailed proof on the number of odd 2-almost primes and odd 3-almost primes less than or equal to n .

By addressing these objectives, this study seeks to contribute valuable insights into the role of odd k -almost primes in the broader context of number theory and their implications for prime counting functions.

Chapter 2

Preliminaries and Some Known Results

This chapter consists of definitions and some known results that are useful in understanding the main context of this study.

2.1 Preliminaries

This section consists of the definitions that will be useful in this study.

Definition 2.1. [1] The set of all **Real Numbers**, denoted by $\mathbb{R}$, are the set of all rational and irrational numbers, and can be represented on a number line.

Definition 2.2. [7] The set of **Natural Numbers** or counting numbers denoted by $\mathbb{N}$, is defined by

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}.$$

Definition 2.3. [4] The **integers** is the set of positive and negative counting numbers together with 0:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

Definition 2.4. [3] Let x be a real number. The **floor function** of x , denoted by $\lfloor x \rfloor$ or floor (x), is defined to be the greatest integer that is less than or equal to x .

Example 2.1. Let $x = 5.8$. Then,

$$\lfloor 5.8 \rfloor = 5$$

This is because 5 is the greatest integer less than or equal to 5.8. Similarly, for a negative value, let $x = -3.4$:

$$\lfloor -3.4 \rfloor = -4$$

Here, -4 is the greatest integer less than or equal to -3.4 .

Definition 2.5. [4] An integer n is **even** if there exists an integer k such that $n = 2k$.

An integer n is **odd** if there exists an integer k such that $n = 2k + 1$.

Definition 2.6. [2] An integer $p > 1$ is called a **prime number**, or simply a prime, if its only positive divisors are 1 and p .

Example 2.2. The integers 2, 3, 5, 7, 11, 13, 17, 19 are prime integers, since each of these integers consists of only two factors: 1 and itself.

Definition 2.7. [2] An integer greater than 1 that is not a prime is termed **composite**.

Example 2.3. The integers 4, 6, 8, 9, 10, 12, 14, 15, and 16 are composite integers, since each of these integers has factors other than 1 and itself.

Definition 2.8. [2] A **prime factor** is a factor that is prime, that is, one that cannot itself be factored. In general, a prime factorization takes the form

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where p_i are prime factors and α_i are their orders.

Example 2.4. Table 2.1 lists the prime factorizations of the integers from 1 to 10. Each number is expressed as the product of prime numbers, with 1 having no prime factors. Prime numbers such as 2, 3, 5, and 7 are listed as themselves, while composite numbers are broken down into their prime factors, such as $4 = 2^2$, $6 = 2 \cdot 3$, and $9 = 3^2$.

Table 2.1: Prime factorizations of the integers from 1 to 10

n	Prime Factorization
1	None
2	2
3	3
4	2^2
5	5
6	$2 \cdot 3$
7	7
8	2^3
9	3^2
10	$2 \cdot 5$

Definition 2.9. [9] An **odd prime** is a prime number p such that $p > 2$ and p is an odd integer. The only **even prime** is 2, all prime numbers other than 2 are classified as odd primes.

Definition 2.10. [4] The **Prime Counting Function** $\pi(x)$ is defined, for all positive integers n , as the number of primes less than or equal to n ; that is,

$$\pi(n) = |\{p \leq n \mid p \text{ is prime}\}|$$

$\Pi(n)$ is defined as the number of *odd primes* less than or equal to n . Since the only even prime is 2, this can be expressed as:

$$\Pi(n) = \pi(n) - 1.$$

Example 2.5. The prime counting function $\pi(n)$ counts how many prime numbers exist that are less than or equal to n .

For instance, when $n = 5$, the primes less than or equal to 5 are 2, 3, and 5, resulting in a count of 3 primes. Thus, $\pi(5) = 3$. This illustrates that there are three primes less than or equal to 5.

Similarly, when $n = 8$, the primes less than or equal to 8 are 2, 3, 5, and 7. In this case,

there are four primes, meaning that $\pi(8) = 4$. This shows that there are four primes less than or equal to 8.

Table 2.2 provides a list of prime numbers less than or equal to a given integer n and the corresponding values of the prime counting function $\pi(n)$.

Table 2.2: Prime numbers less than or equal to n and corresponding prime counting function $\pi(n)$ values

n	Primes less than or equal to n	$\pi(n)$
2	2	1
3	2 and 3	2
4	2 and 3	2
5	2, 3, and 5	3
6	2, 3, and 5	3
7	2, 3, 5, and 7	4
8	2, 3, 5, and 7	4

Definition 2.11. [6] If a number is the product of exactly k prime numbers, that are the same or distinct, it is said to be ***k-almost prime***

Example 2.6. Prime numbers are considered as ***1-almost Primes*** since each has exactly 1 prime factor.

The integers 4, 6, and 9 are examples of ***2-almost Primes Numbers*** since they have exactly 2 prime factors (not necessarily distinct).

The integers 8, 12, and 18 are examples of ***3-almost Primes Numbers*** since they have exactly 3 prime factors (not necessarily distinct).

The integers 16, 24, and 36 are examples of ***4-almost Primes Numbers*** since they have exactly 4 prime factors (not necessarily distinct).

The integer 32 is a ***5-Almost Prime Number*** since it can be expressed as $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ and has 5 factors, and also 630 is a ***5-Almost Prime Number*** because $630 = 2 \cdot 3 \cdot 3 \cdot 5 \cdot 7$ and has 5 factors.

Table 2.3: Prime Factorizations of k-almost Primes

k-Almost Prime	n	Prime Factorization
2-Almost Prime	4, 6, and 9	$4 = 2 \cdot 2$ $6 = 2 \cdot 3$ $9 = 3 \cdot 3$
3-Almost Prime	8, 12, and 18	$8 = 2 \cdot 2 \cdot 2$ $12 = 2 \cdot 2 \cdot 3$ $18 = 2 \cdot 3 \cdot 3$
4-Almost Prime	16, 24, and 36	$16 = 2 \cdot 2 \cdot 2 \cdot 2$ $24 = 2 \cdot 2 \cdot 2 \cdot 3$ $36 = 2 \cdot 2 \cdot 3 \cdot 3$
5-Almost Prime	32 and 630	$32 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ $630 = 2 \cdot 3 \cdot 3 \cdot 5 \cdot 7$

Definition 2.12. If an odd number is the product of exactly k prime numbers, that are the same or distinct, it is said to be **odd k -almost prime**. This means none of its prime factors can be 2.

Example 2.7. Except for 2, all prime numbers are **odd 1-almost Primes**.

The odd integers 9, 15, and 25 are **odd 2-almost Primes Numbers** since each has exactly 2 prime factors (not necessarily distinct).

The odd integers 27, 75, and 105 are **odd 3-almost Primes Numbers** since each has exactly 3 prime factors (not necessarily distinct).

The odd integers 225, 315, and 525 are **odd 4-almost Primes Numbers** since each has exactly 4 prime factors (not necessarily distinct).

The integer 945 is an **odd 5-Almost Prime Number** since it can be expressed as

$3 \cdot 5 \cdot 7 \cdot 7 \cdot 3$, having exactly 5 prime factors.

Table 2.4 shows the prime factorization of odd k -almost primes as k increases.

Table 2.4: Odd k -Almost Primes

Odd K-ALmost Primes	n	Prime Factorization
Odd 2-Almost Prime	9 and 15	$9 = 3 \cdot 3$ $15 = 3 \cdot 5$
Odd 3-Almost Prime	75 and 105	$75 = 3 \cdot 5 \cdot 5$ $105 = 3 \cdot 5 \cdot 7$
Odd 4-Almost Prime	315 and 525	$315 = 3 \cdot 3 \cdot 5 \cdot 7$ $24 = 3 \cdot 5 \cdot 5 \cdot 7$
Odd 5-Almost Prime	945	$945 = 3 \cdot 3 \cdot 3 \cdot 5 \cdot 7$

Definition 2.13. If an even number is the product of exactly k prime numbers, whether distinct or not, it is said to be an *even k -almost prime*. This means that at least one of its prime factors must be 2.

Example 2.8. The only *even 1-almost Prime* is the integer 2. The integers 4 and 6 are *even 2-Almost Primes* as they have exactly 2 prime factors (not necessarily distinct).

The integers 8 and 12 are *even 3-Almost Primes* as they have exactly 3 prime factors (not necessarily distinct).

As k increases, the factorization complexity grows, as seen in the *even 5-Almost Primes* such as 16 and 32 in Table 2.5 which shows the prime factorizations of several even numbers that are k -almost primes for different values of k .

Table 2.5: Even k -Almost Primes

Type	n	Prime Factorization
Even 1-Almost Prime	2	2
Even 2-Almost Prime	4 and 6	$4 = 2 \cdot 2$ $6 = 2 \cdot 3$
Even 3-Almost Prime	8 and 12	$8 = 2 \cdot 2 \cdot 2$ $12 = 2 \cdot 2 \cdot 3$
Even 4-Almost Prime	16 and 24	$16 = 2 \cdot 2 \cdot 2 \cdot 2$ $24 = 2 \cdot 2 \cdot 2 \cdot 3$
Even 5-Almost Prime	32	$32 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

Definition 2.14. [8] Given a sequence $\{a_n\}_{n=k}^{\infty}$ and numbers m and p satisfying

$k \leq m \leq p$, the summation from m to p of the sequence $\{a_n\}$ is written as

$$\sum_{n=m}^p a_n = a_m + a_{m+1} + \cdots + a_p$$

The variable n is called the **index of summation**. The number m is called the **lower limit of summation**, while the number p is called the **upper limit of summation**.

Example 2.9. Let $a_n = 2n$.

Substitute $n = 3$ into the formula $2n$ and add successive terms until $k = 6$.

$$\begin{aligned}
 \sum_{n=3}^6 2n &= 2(3) + 2(4) + 2(5) + 2(6) \\
 &= 6 + 8 + 10 + 12 \\
 &= 36
 \end{aligned}$$

Definition 2.15. [8] Given two whole numbers n and j with $n \geq j$, the **binomial coefficient** $\binom{n}{j}$ (read, 'n choose j') is the whole number given by

$$\binom{n}{j} = \frac{n!}{j!(n-j)!}$$

Example 2.10. Let $n = 5$ and $j = 2$. Then,

$$\begin{aligned} \binom{5}{2} &= \frac{5!}{2!(5-2)!} \\ &= \frac{5}{2!(3)!} \\ &= \frac{5(4)}{2!} \\ &= \frac{20}{2} \\ &= 10 \end{aligned}$$

2.2 Some Known Results

Theorem 2.16. *Properties of Summation Notation*[8]

Suppose $\{a_n\}$ and $\{b_n\}$ are sequences such that the following sums are defined:

- **Sum and Difference Property:**

$$\sum_{n=m}^p (a_n \pm b_n) = \sum_{n=m}^p a_n \pm \sum_{n=m}^p b_n$$

- **Distributive Property:** For any real number c ,

$$\sum_{n=m}^p ca_n = c \sum_{n=m}^p a_n$$

- **Additive Index Property:** For any natural numbers $m \leq j < j + 1 \leq p$,

$$\sum_{n=m}^j a_n + \sum_{n=j+1}^p a_n = \sum_{n=m}^p a_n$$

- **Re-indexing:** For any integer r ,

$$\sum_{n=m}^p a_n = \sum_{n=m+r}^{p+r} a_{n-r}$$

Lemma 2.17. Summation Formula [8]

•

$$\sum_{k=1}^n c = cn$$

•

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Theorem 2.18. [5] If n is a positive composite number, then n has a prime divisor less than or equal to $\sqrt{n}$.

Chapter 3

Main Results

This chapter provides the iteration on Prime Counting Functions, the iteration on Prime Counting Function using 2-almost primes, and the iteration on Prime Counting Function using odd 2-almost primes. Furthermore, detailed proofs regarding the number of 2-almost primes, odd 2-almost primes, 3-almost primes, and odd 3-almost primes less than or equal to n are provided.

3.1 Iteration on finding the number of primes less than or equal to n

In this section, the procedures in iterative methods for counting prime numbers less than or equal to a given integer n are listed. It begins with traditional prime counting functions, which provide general estimates but may lack precision for smaller values. Next, prime counting functions that utilize 2-almost primes, numbers with exactly two prime factors are explored, offering a more targeted and accurate counting approach. Finally, the refinement achieved by focusing on odd 2-almost primes is considered, further enhancing accuracy. Through this iteration, the assessment of the efficiency and effectiveness of each method contributes to a deeper understanding of prime distribution and its applications.

3.1.1 Iteration on Prime Counting Functions

1. **Make a list.** Start with a list of numbers 2 to n .

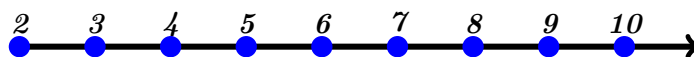
2. **Initial Assumption.** Assume all numbers are potential primes.
3. **Identify the upper limit.** Compute $\sqrt{n}$. This will be the limit for marking multiples of primes
4. **Identify the first prime.** Begin with the first prime number, which is 2
5. **Mark multiples.** Let p be the current prime (initially 2). Mark all multiples of p as 'not prime'. This is because any smaller multiples of p will have been marked by smaller primes. After marking, move to the next number in the list that is still labeled 'prime'. This becomes the new p .
6. **Repeat the Process.** Continue the process of marking multiples for each new prime p . Skip any p that is already labeled as 'not prime'. Stop marking when p exceeds $\sqrt{n}$, because any non-prime number n will have at least one factor less than or equal to $\sqrt{n}$.
7. **Count the primes.** After completing the marking process, count how many numbers in your list are still labeled as primes. This total gives you the number of prime numbers less than or equal to n .

Example 3.1. *Let $n = 10$*

1. **Make a List:** 2, 3, 4, 5, 6, 7, 8, 9, 10

2. **Initial Assumption**

Assume all numbers are potential primes and color them blue.



3. **Identify the upper limit:**

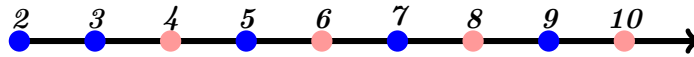
To take the upper limit, take the square root of n . $\sqrt{n} = \sqrt{10} \approx 3.16$. Thus, the upper limit is 3.

4. **Identify the first prime:**

The first prime number is 2.

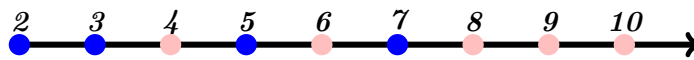
5. **Mark Multiples:**

Mark all multiples of 2 as non prime and color it pink. The multiples of 2 are 4, 6, 8, 10, since $n = 10$.



6. **Repeat the Process:**

The next unmarked number is 3. Mark all multiples of 3 and color it pink. The multiples of 3 are 6 and 9.



Now, since the next prime to consider is 5, greater than $\sqrt{10}$, we stop marking.

7. **Count the Primes:** Count the remaining numbers labeled as primes: 2, 3, 5, 7.

Therefore, there are 4 prime number less than or equal to 10.

3.1.2 Iteration on Prime Counting Function using 2-almost prime

1. **Make a list.** Start with a list of numbers 2 to n .
2. **Initial Assumptions.** Assume all numbers are potential 2-Almost Primes.
3. **Identify the upper limit.** Compute $\sqrt{n}$. This will be useful in determining prime factors.
4. **Identify Primes.** Use method 3.1.1 to find all the prime numbers up to n . Create a list of these primes.
5. **Mark 2-Almost Primes.** For each pair of primes (p, q) from the list in step 4, not necessarily distinct;

- a) Calculate the product pq .
- b) If $pq \leq n$ mark it as 2-almost prime.

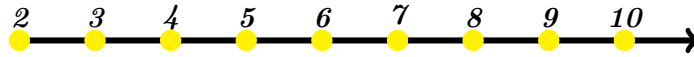
6. **Count the 2-Almost Primes.** After marking, count how many numbers in the list are labeled as 2-almost primes. This total gives you the number of 2-almost primes less than or equal to n .

Example 3.2. Let $n = 10$

1. **Make a List:** 2, 3, 4, 5, 6, 7, 8, 9, 10

2. **Initial Assumptions:**

Assume that all numbers are potential 2-almost primes and color them yellow.



3. **Identify the upper limit:**

To find the upper limit, take the square root of n . $\sqrt{n} = \sqrt{10} \approx 3.16$.

4. **Identify Primes less than or equal to n :**

The primes less than or equal to 10 are 2, 3, 5, and 7.

5. **Mark 2-Almost Primes:**

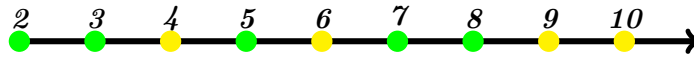
Calculate the product pq , where p and q are primes from step 4. If $pq \leq n$, mark it as 2-almost prime. Let $p = 2$,
 q can be substituted as 2, 3, 5, and 7

p	q	pq	$pq \leq n$
2	2	4	2-Almost Prime
2	3	6	2-Almost Prime
2	5	10	2-Almost Prime
2	7	14	Product is greater than n

Hence, 4, 6, and 10 are the 2-almost primes less than or equal to n . Let $p = 3$, q can be substituted as 2, 3, 5, and 7

p	q	pq	$pq \leq n$
3	2	6	2-Almost Prime
3	3	9	2-Almost Prime
3	5	15	Product is greater than n
3	7	21	Product is greater than n

Hence, 6 and 9 are the 2-almost primes less than or equal to n .



6. Count the 2-Almost Primes:

The 2-Almost Primes less than or equal to 10 are: 4, 6, 9, and 10. Therefore, there are 4 of 2-Almost Primes less than or equal to 10.

3.1.3 Iteration on Prime Counting Function using Odd 2-almost prime

1. **Make a list.** Start with a list of numbers 2 to n .
2. **Initial Assumptions.** Assume all numbers are potential Odd 2-Almost Primes, except for 2. Set the status of 2 to “not Odd 2-Almost Prime” .
3. **Identify the upper limit.** Compute $\sqrt{n}$. This will be useful for determining prime factors.
4. **Identify Odd Primes.** Use method 3.1.1 to find all the odd prime numbers up to n . Create a list of these primes.
5. **Mark Odd 2-Almost Primes.** For each pair of primes (p, q) from the list in step 4 not necessarily distinct;

- a) Calculate the product pq .
- b) If $pq \leq n$ mark it as Odd 2-Almost Prime.

6. **Count the Odd 2-Almost Primes.** After marking, count how many number in your list are labeled as Odd 2-Almost Primes. This total gives you the number of Odd 2-Almost Primes less than or equal to n .

Example 3.3. Let $n = 10$

1. **Make a List:** 2, 3, 4, 5, 6, 7, 8, 9, 10

2. **Initial Assumptions:**

Mark 2 in color pink as not odd 2-almost prime, and color the remaining numbers blue as potential Odd 2-almost primes



3. **Identify the upper limit:** To take the upper limit, take the square root of n .

$$\sqrt{n} = \sqrt{10} \approx 3.16 .$$

4. **Identify Odd Primes:**

The odd primes less than or equal to 10 are 3, 5, and 7.

5. **Mark Odd 2-Almost Primes:**

Calculate the product pq , where p and q are primes from step 5, such that $pq \leq n$.

If $pq \leq n$, mark it odd 2-almost prime. Let $p = 3$,

q can be substituted as 3, 5 and 7

p	q	pq	$pq \leq n$
3	3	9	Odd 2-Almost Prime
3	5	15	Product is greater than n
2	7	21	Product is greater than n

Hence, 9 is the only odd 2-almost prime less than or equal to n .



6. Count the Odd 2-Almost Primes:

The Odd 2-Almost Primes less than or equal to 10 is only 9. Therefore, there is only 1 Odd 2-Almost Prime less than or equal to 10.

3.2 2-Almost Primes and Odd 2-Almost Primes

3.2.1 Prime counting functions using 2-almost Primes

Theorem 3.1. For any natural number n , the number of 2-almost primes is given by

$$\pi^{(2)}(n) = \sum_{i=1}^{\pi(n^{\frac{1}{2}})} \left[\pi\left(\frac{n}{p_i}\right) - i + 1 \right] \quad (3.1)$$

where p_i is the i th prime, $\pi(n)$ is the prime counting function and $\pi^{(k)}(n)$ denotes k -almost prime function.

Proof. Suppose n is a natural number, and $u \leq n$ is a 2-almost prime where $u = pq$, p and q are primes (not necessarily distinct).

Consider the set of primes $p_1, p_2, \dots, p_i$ such that $p_i \leq \sqrt{n}$; that is, $i \leq \pi(n^{\frac{1}{2}})$. For each prime p_i , count the number of primes q such that the product $u = pq$ satisfies $u \leq n$.

This condition can be rewritten as:

$$p_i q \leq n \quad \Rightarrow \quad q \leq \frac{n}{p_i}.$$

Then, count the number of primes such that $q \leq \frac{n}{p_i}$, which is given by $\pi\left(\frac{n}{p_i}\right)$. The number of primes less than p_i is exactly $i - 1$, since $p_1, p_2, \dots, p_{i-1}$ are all the primes

less than p_i . Thus, the number of valid primes q for a given p_i , where $p_i \leq q$, is obtained by subtracting the number of primes smaller than p_i from $\pi\left(\frac{n}{p_i}\right)$. This gives the count:

$$\pi\left(\frac{n}{p_i}\right) - (i - 1).$$

This implies that we only count pairs (p_i, q) such that $p_i \leq q$, thereby preventing any pairs from being counted more than once.

For $i = 1$:

$$\pi\left(\frac{n}{p_1}\right) - (1 - 1) = \pi\left(\frac{n}{p_1}\right)$$

This counts the number of primes q such that $p_1 q \leq n$ where p_1 is the first prime.

For $i = 2$:

$$\pi\left(\frac{n}{p_2}\right) - (2 - 1) = \pi\left(\frac{n}{p_2}\right) - 1$$

This counts the number of primes q such that $p_2 q \leq n$ where p_2 is the second prime.

For $i = 3$:

$$\pi\left(\frac{n}{p_3}\right) - (3 - 1) = \pi\left(\frac{n}{p_3}\right) - 2$$

This counts the number of primes q such that $p_3 q \leq n$ where p_3 is the third prime.

Summing over all primes $p_1, p_2, \dots, p_i$, the total number of 2-almost primes is:

$$\sum_{i=1}^{\pi(n^{\frac{1}{2}})} \left[\pi\left(\frac{n}{p_i}\right) - (i - 1) \right].$$

□

Example 3.4. The number of 2-almost primes up to n is,

$$\pi^{(2)}(n) = \sum_{i=1}^{\pi(n^{\frac{1}{2}})} \left[\pi\left(\frac{n}{p_i}\right) - i + 1 \right]$$

Let $n=150$ and $\pi(n^{\frac{1}{2}}) = \pi(150^{\frac{1}{2}}) = \pi(12.25) = 5$

$$\begin{aligned}
\therefore \pi^{(2)}(150) &= \sum_{i=1}^5 \left[\pi \left(\frac{150}{p_i} \right) - i + 1 \right] \\
&= [\pi \left(\frac{150}{2} \right) - 1 + 1] + [\pi \left(\frac{150}{3} \right) - 2 + 1] + [\pi \left(\frac{150}{5} \right) - 3 + 1] + [\pi \left(\frac{150}{7} \right) - 4 + 1], \\
&\quad + [\pi \left(\frac{150}{11} \right) - 5 + 1], \\
&= [\pi(75) - 0] + [\pi(50) - 1] + [\pi(30) - 2] + [\pi(21.43) - 3] + [\pi(13.64) - 4], \\
&= (21 - 0) + (15 - 1) + (10 - 2) + (8 - 3) + (6 - 4), \\
&= 21 + 14 + 8 + 5 + 2, \\
&= 50.
\end{aligned}$$

Therefore, there are 50 pairs of 2-almost primes up to 150. The table below shows all the prime pairs (p, q) , and their corresponding 2-almost primes pq for primes less than or equal to 150.

Table 3.1: List of all 2-almost primes less than or equal to 150

p	q	pq	p	q	pq	p	q	pq	p	q	pq	p	q	pq
2	2	4	2	31	62	2	73	146	3	31	93	5	19	95
2	3	6	2	37	74	3	3	9	3	37	111	5	23	115
2	5	10	2	41	82	3	5	15	3	41	123	5	29	145
2	7	14	2	43	86	3	7	21	3	43	129	7	7	49
2	11	22	2	47	94	3	11	33	3	47	141	7	11	77
2	13	26	2	53	106	3	13	39	5	5	25	7	13	91
2	17	34	2	59	118	3	17	51	5	7	35	7	17	119
2	19	38	2	61	122	3	19	57	5	11	55	7	19	133
2	23	46	2	67	134	3	23	69	5	13	65	11	11	121
2	29	58	2	71	142	3	29	87	5	17	85	11	13	143

3.2.2 Prime counting functions using odd 2-almost Primes

Theorem 3.2. *Odd 2-Almost Prime*

For any natural number n , the number of odd 2-almost primes is given by

$$\Pi^{(2)}(n) = \sum_{i=1}^t \Pi\left(\frac{n}{p_{i+1}}\right) - \frac{t(t-1)}{2}, \quad (3.2)$$

where $t=\Pi(n^{1/2})$ and p_i the i th prime number

Proof. Suppose n is a natural number, $u \leq n$ is an odd 2-almost prime where $u = pq$, p and q are odd primes (not necessarily distinct).

Let $p = 3$, the smallest odd prime. Then, for $u = pq$, it follows that:

$$u = 3q$$

where q is another odd prime. Hence, the constraint is:

$$3q \leq n \Rightarrow q \leq \frac{n}{3}.$$

Thus, q must be an odd prime number. The function $\Pi(n)$ represents the number of odd primes less than or equal to n . Since $q \leq \frac{n}{3}$, the function $\Pi(\frac{n}{3})$ counts all odd primes that are less than or equal to $\frac{n}{3}$. For each odd prime q that satisfies $q \leq \frac{n}{3}$, $u = 3q$ is an odd 2-almost prime. The number of such odd primes q is exactly $\Pi(\frac{n}{3})$, consequently, $\Pi(\frac{n}{3})$ is the total number of odd 2-almost primes whose only factor is 3.

Next, repeat the same process for the next odd prime, $p = 5$. Then, for $u = pq$, it follows that $u = 5q$, where q is another odd prime. The constraint is:

$$5q \leq n \Rightarrow q \leq \frac{n}{5}.$$

The function $\Pi(\frac{n}{5})$ is the total number of odd 2-almost primes with 5 as one of the prime factors.

Next, fix $p = 7$ and q varying over the primes to $\frac{n}{7}$. Subsequently, $\Pi(\frac{n}{7})$ gives the total number of odd 2-almost primes where 7 is one of the prime factors. Continue this process for each subsequent odd prime p_i until $\frac{n}{p_{i+1}} \leq 3$ (the smallest odd prime number).

All odd 2-almost primes obtained for each prime p_i are added. Since primes larger than $\sqrt{n}$ are not considered (because their products would exceed n), the product of two primes greater than $\sqrt{n}$ results in a number greater than n . Thus, $t = \Pi(n^{1/2})$ is defined, where $\Pi(n)$ is the prime counting function. This means t represents the number of odd primes up to $\sqrt{n}$, which are used to form the 2-almost primes. Hence, the sum of all odd 2-almost primes is:

$$\sum_{i=1}^t \Pi\left(\frac{n}{p_{i+1}}\right)$$

To avoid over-counting, the number of duplicated counts is subtracted. Each number may be counted more than once when considering different primes. This over-counting occurs for any pair of primes, so the binomial coefficient $\binom{t}{2}$, which accounts for choosing two primes from t primes, is subtracted. This gives:

$$-\binom{t}{2} = -\frac{t(t-1)}{2}$$

Therefore,

$$\Pi^{(2)}(n) = \sum_{i=1}^t \Pi\left(\frac{n}{p_{i+1}}\right) - \frac{t(t-1)}{2}$$

□

Proposition 3.3. *Let n be any natural number,*

$$\Pi^{(2)}(n) = \pi^{(2)}(n) - \pi\left(\frac{n}{2}\right). \quad (3.3)$$

Proof. A 2-almost prime pq can either be an odd 2-almost primes ($\Pi^{(2)}(n)$) or an even 2-almost primes. Thus, the total number of 2-almost primes up to n , denoted as $\pi^{(2)}(n)$, can be expressed as:

$$\pi^{(2)}(n) = \Pi^{(2)}(n) + (\text{number of even 2-almost primes up to } n).$$

An even 2-almost prime takes the form $2p$, where p is an even or odd prime. For $2p \leq n$, $p \leq \frac{n}{2}$. The number of such odd primes p is exactly, $\pi\left(\frac{n}{2}\right)$, which counts all primes less than or equal to $\frac{n}{2}$. Thus, the number of even 2-almost primes is $\pi\left(\frac{n}{2}\right)$.

Substituting the count of even 2-almost primes into the formula for $\pi^{(2)}(n)$,

$$\pi^{(2)}(n) = \Pi^{(2)}(n) + \pi\left(\frac{n}{2}\right).$$

Rearranging this equation to isolate $\Pi^{(2)}(n)$, the count of odd 2-almost primes is

$$\Pi^{(2)}(n) = \pi^{(2)}(n) - \pi\left(\frac{n}{2}\right).$$

□

3.3 Algebraic proof of odd 2-almost primes

Proving algebraically for odd 2-almost primes requires modifying the original formula for 2-almost primes, which includes even numbers. To focus specifically on odd 2-almost primes, all even numbers must be excluded. This ensures that only valid odd 2-almost primes are counted. This refinement is necessary for two key reasons. First, it guarantees accuracy by ensuring that only odd numbers meeting the criteria for 2-almost primes are included. Second, it prevents over-counting, which can occur due to the presence of the

prime number 2. Since even numbers always have 2 as a factor, they would otherwise be mistakenly counted as odd 2-almost primes. By constructing an algebraic proof that specifically targets odd 2-almost primes, it becomes possible to correctly identify and count these numbers. This adjustment eliminates errors introduced by the inclusion of even numbers and provides a clearer, more rigorous understanding of the properties and distribution of odd 2-almost primes.

Proof.

$$\begin{aligned}
\pi^{(2)}(n) &= \sum_{i=1}^{\pi(n^{\frac{1}{2}})} \left[\pi\left(\frac{n}{p_i}\right) - i + 1 \right] \\
&= \sum_{i=1}^r \left[\pi\left(\frac{n}{p_i}\right) - i + 1 \right], \quad \text{where } r = \pi(n^{\frac{1}{2}}) \\
&= \sum_{i=1}^r \pi\left(\frac{n}{p_i}\right) - \sum_{i=1}^r i + \sum_{i=1}^r 1 \\
&= \sum_{i=1}^r \pi\left(\frac{n}{p_i}\right) - \frac{r(r+1)}{2} + r \\
&= \sum_{i=1}^r \pi\left(\frac{n}{p_i}\right) - \frac{r(r+1)}{2} + \frac{2r}{2} \\
&= \sum_{i=1}^r \pi\left(\frac{n}{p_i}\right) - \frac{r^2 + r}{2} + \frac{2r}{2} \\
&= \sum_{i=1}^r \pi\left(\frac{n}{p_i}\right) - \frac{r^2 + r - 2r}{2} \\
&= \sum_{i=1}^r \pi\left(\frac{n}{p_i}\right) - \frac{r^2 - r}{2} \\
&= \sum_{i=1}^r \pi\left(\frac{n}{p_i}\right) - \frac{r(r-1)}{2}.
\end{aligned} \tag{3.4}$$

By Definition 2.10,

$$\Pi(n^{\frac{1}{2}}) = \pi(n^{\frac{1}{2}}) - 1$$

$$\Pi(n^{\frac{1}{2}}) + 1 = \pi(n^{\frac{1}{2}})$$

$$\Pi(n^{\frac{1}{2}}) + 1 = r, \quad \text{since } r = \pi(n^{\frac{1}{2}})$$

Now, let $\Pi(n^{\frac{1}{2}}) = t$,

$$t + 1 = r$$

Substituting this value of r into equation (3.4), the result is:

$$\begin{aligned}\pi^{(2)}(n) &= \sum_{i=1}^{t+1} \pi\left(\frac{n}{p_i}\right) - \frac{(t+1)[(t+1)-1]}{2} \\ &= \sum_{i=1}^{t+1} \pi\left(\frac{n}{p_i}\right) - \frac{t(t+1)}{2}\end{aligned}$$

Using the relationship $\pi\left(\frac{n}{p_i}\right) = \Pi\left(\frac{n}{p_i}\right) + 1$, it follows that:

$$\begin{aligned}&= \sum_{i=1}^{t+1} \left[\Pi\left(\frac{n}{p_i}\right) + 1 \right] - \frac{t(t+1)}{2}, \\ &= \sum_{i=1}^{t+1} \left[\Pi\left(\frac{n}{p_i}\right) \right] + \sum_{i=1}^{t+1} 1 - \frac{t(t+1)}{2}, \\ &= \sum_{i=1}^{t+1} \left[\Pi\left(\frac{n}{p_i}\right) \right] + (t+1) - \frac{t(t+1)}{2}, \\ &= \sum_{i=1}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{2(t+1)}{2} - \frac{t(t+1)}{2}, \\ &= \sum_{i=1}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{2t+2}{2} - \frac{t^2+t}{2}, \\ &= \sum_{i=1}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{2t+2-t^2-t}{2}, \\ &= \sum_{i=1}^{t+1} \Pi\left(\frac{n}{p_i}\right) - \frac{t^2-t-2}{2} \\ &= \sum_{i=1}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{(t+1)(t-2)}{2}\end{aligned}\tag{3.5}$$

Substituting (3.5) into (3.3), we get

$$\begin{aligned}&= \sum_{i=1}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{(t+1)(2-t)}{2} - \pi\left(\frac{n}{2}\right), \\ &= \Pi\left(\frac{n}{p_1}\right) + \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{(t+1)(2-t)}{2} - \pi\left(\frac{n}{2}\right)\end{aligned}$$

From $\Pi(n) = \pi(n) - 1$, implies $\Pi(\frac{n}{p_1}) = \pi(\frac{n}{p_1}) - 1$. Hence,

$$\begin{aligned}
&= \pi\left(\frac{n}{p_1}\right) - 1 + \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{(t+1)(2-t)}{2} - \pi\left(\frac{n}{2}\right), \\
&= \pi\left(\frac{n}{2}\right) - 1 + \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{(t+1)(2-t)}{2} - \pi\left(\frac{n}{2}\right), \text{ since } p_1 = 2 \\
&= \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{(t+1)(2-t)}{2} - 1, \\
&= \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{(t+1)(2-t)}{2} - \frac{2}{2}, \\
&= \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{2t - t^2 + 2 - t}{2} - \frac{2}{2}, \\
&= \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{-t^2 + t + 2}{2} - \frac{2}{2}, \\
&= \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{-t^2 + t + 2 - 2}{2}, \\
&= \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) + \frac{-t^2 + t}{2}, \\
&= \sum_{i=2}^{t+1} \Pi\left(\frac{n}{p_i}\right) - \frac{t(t-1)}{2}, \\
&= \sum_{i=1}^t \Pi\left(\frac{n}{p_{i+1}}\right) - \frac{t(t-1)}{2}.
\end{aligned}$$

Therefore,

$$\Pi^{(2)}(n) = \sum_{i=1}^t \Pi\left(\frac{n}{p_{i+1}}\right) - \frac{t(t-1)}{2}.$$

□

Example 3.5. The number of odd 2-almost primes up to n is,

$$\Pi^{(2)}(n) = \sum_{i=1}^t \Pi\left(\frac{n}{p_{i+1}}\right) - \frac{t(t-1)}{2}$$

Let $n = 150$, and $t = \Pi(150^{\frac{1}{2}}) = \Pi(12.25) = 4$

$$\begin{aligned}
\therefore \Pi^{(2)}(150) &= \sum_{i=1}^4 \Pi\left(\frac{150}{p_{i+1}}\right) - 6, \\
&= \Pi\left(\frac{150}{3}\right) + \Pi\left(\frac{150}{5}\right) + \Pi\left(\frac{150}{7}\right) + \Pi\left(\frac{150}{11}\right) - 6, \\
&= \Pi(50) + \Pi(30) + \Pi(21.43) + \Pi(13.64) - 6, \\
&= 14 + 9 + 7 + 5 - 6, \\
&= 29.
\end{aligned}$$

Therefore, there are 29 odd 2-almost primes up to 150. Table 3.2 shows all the prime pairs p, q , and their corresponding odd 2-almost primes pq for primes less than or equal to 150.

Table 3.2: List of all odd 2-almost primes less than or equal to 150

p	q	pq	p	q	pq	p	q	pq
3	3	9	3	37	111	5	23	115
3	5	15	3	41	123	5	29	145
3	7	21	3	43	129	7	7	49
3	11	33	3	47	141	7	11	77
3	13	39	5	5	25	7	13	91
3	17	51	5	7	35	7	17	119
3	19	57	5	11	55	7	19	133
3	23	69	5	13	65	11	11	121
3	29	87	5	17	85	11	13	143
3	31	93	5	19	95			

3.4 3-Almost Primes and Odd 3-Almost Primes

3.4.1 Prime counting functions using 3-almost Primes

Theorem 3.4. *For any natural number n , the number of 3-almost primes is given by*

$$\pi^{(3)}(n) = \sum_{i=1}^{\pi(n^{\frac{1}{3}})} \sum_{j=i}^{\pi\left(\sqrt{\frac{n}{p_i}}\right)} \left[\pi\left(\frac{n}{p_i p_j}\right) - j + 1 \right] \quad (3.4)$$

where p_j is the j -th prime, $\pi(n)$ is the prime counting function, and $\pi^{(k)}(n)$ denotes the k -almost prime function.

Proof. Suppose n is a natural number, and $u \leq n$ is a 3-almost prime where $u = pp_jq$, p, p_j, q are primes satisfying the non-decreasing order $p \leq p_j \leq q$.

Case 1: $p = p_j = q$

Consider the set of primes $p_1, p_2, \dots, p_i$ such that $p_i = p_j = q$. Since $u \leq n$, this condition can be rewritten as:

$$p_i^3 \leq n \Rightarrow p_i \leq n^{\frac{1}{3}}.$$

Then, count the number of primes p_i such that $p_i \leq n^{\frac{1}{3}}$, which is given by $\pi(n^{\frac{1}{3}})$.

Summing over all possible p_i values, the result is

$$\sum_{i=1}^{\pi(n^{\frac{1}{3}})} 1.$$

Case 2: $p_i = p_j \leq q$

$$p_i^2 q \leq n \Rightarrow p_i^2 = \frac{n}{q}.$$

However, since p_i is repeated, the smallest factor must satisfy

$$p_i^2 \leq n$$

Taking the square root

$$p_i \leq n^{\frac{1}{2}}.$$

Then, count the number of primes $p_i = \pi(n^{\frac{1}{2}})$ such that $p_i \leq n^{\frac{1}{2}}$. For each fixed $p_i = p_1$, the third prime factor q must satisfy

$$p_i^2 q \leq n \Rightarrow q \leq \frac{n}{p_i^2}$$

The number of such valid q is $\pi(\frac{n}{p_i^2})$, but since $p_i^2 \leq q$, primes smaller than p_i must be excluded. The number of primes p_i is exactly $i - 1$, since $p_1, p_2, \dots, p_{i-1}$. Hence, the number of valid primes q for a given $p_i = p_j$, where $p_i = p_j \leq q$, is obtained by subtracting the number of primes smaller than p_i^2 from $\pi(\frac{n}{p_i^2})$. This gives the count:

$$\pi\left(\frac{n}{p_i^2}\right) - (i - 1)$$

Summing over all possible values of p_i ,

$$\sum_{i=1}^{\pi(n^{\frac{1}{2}})} \left[\pi\left(\frac{n}{p_i^2}\right) - (i - 1) \right]$$

Case 3: $p_i \leq p_j = q$

$$p_i \leq p_j = q \Rightarrow p_i p_j^2 \leq n \Rightarrow p_i \leq \left(\frac{n}{p_1^2}\right).$$

However, since $p_i \leq p_j$, the smallest factor p_i must satisfy $p_i \leq n^{\frac{1}{3}}$ because if p_i is larger than $n^{\frac{1}{3}}$, then even for $p_j = p_i$,

$$p_i p_j^2 \leq n = p_i^3 \leq n.$$

Hence, the number of valid primes p_i is $\pi(n^{\frac{1}{3}})$. Summing over all values of p_i , the result is

$$\sum_{i=1}^{\pi(n^{\frac{1}{3}})}.$$

For each fixed p_i , the prime p_1 must satisfy

$$p_i p_j^2 \leq n.$$

Rearranging for p_j

$$p_j \leq \left(\frac{n}{p_i}\right)^{\frac{1}{2}}.$$

The number of such primes p_j is $\pi\left(\sqrt{\frac{n}{p_i}}\right)$. However, since p_j must be at least as large as p_i , primes smaller than p_i are excluded. The number of primes smaller than p_i is $i - 1$, so the number of valid p_j values is

$$\pi\left(\sqrt{\frac{n}{p_i}}\right) - (i - 1)$$

Summing over all p , the count is

$$\sum_{i=1}^{\pi(n^{\frac{1}{3}})} \pi\left(\sqrt{\frac{n}{p_i}}\right) - (i - 1).$$

Case 4: $p_i \leq p_j \leq q$

Since $p_i \leq p_j \leq q \leq n$, the largest possible value for p_i occurs when $p_i = p_j = q$, this

condition can be rewritten as

$$p_i^3 \leq n \Rightarrow p_i \leq n^{\frac{1}{3}}.$$

Thus, p_i can be any primes less than or equal to $n^{\frac{1}{3}}$, and the number of such primes p_i is $\pi(n^{\frac{1}{3}})$. Hence,

$$\sum_{i=1}^{\pi(n^{\frac{1}{3}})}.$$

For each fixed p_i , $p_j q$ must satisfy

$$p_i p_j q \leq n \Rightarrow p_j q \leq \frac{n}{p_i}.$$

Since $p_j \geq p_i$, the largest possible value for p_j is

$$p_j \leq \left(\frac{n}{p_i} \right)^{\frac{1}{2}}$$

Summing over all possible p_j values:

$$\sum_{j=i}^{\pi\left(\sqrt{\frac{n}{p_i}}\right)}.$$

For each valid pair (p_j, p_i) , the third prime factor q must satisfy

$$p_j p_i q \leq n \Rightarrow q \leq \frac{n}{p_j p_i}.$$

The number of valid primes q is $\pi\left(\frac{n}{p_j p_i}\right)$. However, since $p_i \leq q$, primes smaller than p_i must be excluded. The number of such primes is exactly $i - 1$. Thus, the count for q is

$$\pi\left(\frac{n}{p_j p_i}\right) - (i - 1).$$

Therefore,

$$\pi^{(3)}(n) = \sum_{i=1}^{\pi(n^{\frac{1}{3}})} \sum_{j=i}^{\pi\left(\sqrt{\frac{n}{p_j}}\right)} \left[\pi\left(\frac{n}{p_j p_i}\right) - i + 1 \right].$$

□

Example 3.6. *The number of 3-almost primes up to n is,*

$$\pi^{(3)}(n) = \sum_{i=1}^{\pi(n^{\frac{1}{3}})} \sum_{j=1}^{\pi\left(\sqrt{\frac{n}{p_i}}\right)} \left[\pi\left(\frac{n}{p_i p_j}\right) - j + 1 \right]$$

Let $n = 150$, and $\pi(n^{\frac{1}{3}}) = \pi(150^{\frac{1}{3}}) = \pi(5.313) = 3$, implies that there are 3 primes less than or equal to 5.313. Thus, the outer sum runs over $i = 1, 2, 3$ corresponding to $p_1 = 2, p_2 = 3$, and $p_3 = 5$.

Evaluate $\pi\left(\frac{n}{p_i}\right)^{\frac{1}{2}}$ for $i = 1, 2, 3$.

For $p_1 = 2$

$$\pi\left(\sqrt{\frac{n}{p_i}}\right) = \pi\left(\sqrt{\frac{150}{2}}\right) = \pi(8.66) = 4$$

For $p_2 = 3$

$$\pi\left(\sqrt{\frac{n}{p_i}}\right) = \pi\left(\sqrt{\frac{150}{3}}\right) = \pi(7.07) = 4$$

For $p_3 = 5$

$$\pi\left(\sqrt{\frac{n}{p_i}}\right) = \pi\left(\sqrt{\frac{150}{5}}\right) = \pi(5.48) = 3$$

Thus, the inner sum now starts from $j = i$, implies

a) $j = 1, 2, 3, 4$ for $i = 1$

b) $j = 2, 3, 4$ for $i = 2$

c) $j = 3$ for $i = 3$

Table 3.3: Prime numbers less than or equal to n and corresponding prime counting function $\pi(n)$ values

n	Primes less than or equal to n	$\pi(n)$
37	2, 3, 5, 7, 11, 13 ,17 ,19 ,23 ,29 , 31, and 37	12
25	2, 3, 5, 7, 11, 13, 17, 19, and 23	9
16	2, 3, 5, 7, 11, and 13	6
10	2, 3, 5, and 7	4
7	2, 3, 5, and 7	4
6	2, 3, and 5	3

$$\begin{aligned}
\therefore \pi^{(3)}(150) &= \sum_{j=1}^4 \left[\pi \left(\frac{150}{2p_j} \right) - j + 1 \right] + \sum_{j=2}^4 \left[\pi \left(\frac{150}{3p_j} \right) - j + 1 \right] + \sum_{j=3}^4 \left[\pi \left(\frac{150}{5p_j} \right) - j + 1 \right] \\
&= \left[\left[\pi \left(\frac{150}{2(2)} \right) - 1 + 1 \right] + \left[\pi \left(\frac{150}{2(3)} \right) - 2 + 1 \right] + \left[\pi \left(\frac{150}{2(5)} \right) - 3 + 1 \right] \right. \\
&\quad \left. + \left[\pi \left(\frac{150}{2(7)} \right) - 4 + 1 \right] \right] + \left[\left[\pi \left(\frac{150}{3(3)} \right) - 2 + 1 \right] + \left[\pi \left(\frac{150}{3(5)} \right) - 3 + 1 \right] \right. \\
&\quad \left. + \left[\pi \left(\frac{150}{3(7)} \right) - 4 + 1 \right] \right] + \left[\pi \left(\frac{150}{5(5)} \right) - 3 + 1 \right] \\
&= [\pi(37.5) + (\pi(25) - 1) + (\pi(15) - 2) + (\pi(10.71) - 3)] \\
&\quad + [\pi(16.67) - 1] + (\pi(10) - 2) + (\pi(7.14) - 3) + [\pi(6) - 2] \\
&= [12 + (9 - 1) + (6 - 2) + (4 - 3)] + [(6 - 1) + (4 - 2) + (4 - 3)] + (3 - 2) \\
&= [12 + 8 + 4 + 1] + [5 + 2 + 1] + 1 \\
&= 25 + 8 + 1 \\
&= 34
\end{aligned}$$

Therefore, there are 34 of 3-almost primes that are less than or equal to 150, and these primes are listed in Table 3.4.

Table 3.4: List of all 3-almost primes less than or equal to 150

p	q	r	pq	p	q	r	pq	p	q	r	pq
2	2	2	8	2	2	31	124	2	5	5	50
2	2	3	12	2	2	39	148	2	5	7	70
2	2	5	20	2	3	3	18	2	5	11	110
2	2	7	28	2	3	5	30	2	5	13	130
2	2	11	44	2	3	7	42	2	7	7	98
2	2	13	52	2	3	11	66	3	3	3	27
2	2	17	68	2	3	13	78	3	3	5	45
2	2	19	76	2	3	17	102	3	3	7	63
2	2	23	92	2	3	19	114	3	3	11	99
2	2	29	116	2	3	23	138	3	3	13	117

p	q	r	pq
3	5	5	75
3	5	7	105
3	7	7	147
5	5	5	125

3.4.2 Odd 3-almost Primes

Theorem 3.5. *Odd 3-Almost Prime*

For any natural number n , the number of odd 3-almost primes is given by

$$\Pi^3(n) = \sum_{j=2}^{s-1} \left[\sum_{i=1}^t \Pi_{p_j} \left(\frac{n}{p_j p_{i+j-1}} \right) - \frac{t(t-1)}{2} \right] \quad (3.5)$$

where $t = \Pi_{p_j} \left(\left\lfloor \frac{n}{p_j} \right\rfloor^{\frac{1}{2}} \right)$, s is the least value for which $t = 0$, $\Pi_k(n)$ is the number of primes greater than or equal to k and less than or equal to n , and p_j is the j th prime.

Proof. Suppose n is a natural number, and $u \leq n$ is an odd 3-almost primes, where $u = pp_iq$, p , p_i , and q are odd primes (not necessarily distinct) and $p \leq p_i \leq q$.

Let $p = 3$, the smallest odd prime, then

$$u = 3p_i q$$

where p_i and q are odd primes. The constraint follows as:

$$3p_i q \leq n \Rightarrow p_i q \leq \frac{n}{3}.$$

Thus, $p_i q$ must be at most $\frac{n}{3}$.

For each fixed p_i , the number of valid choices for q must satisfy:

$$q \leq \frac{n}{3p_i}.$$

The function $\Pi_{p_i} \left(\frac{n}{3p_i} \right)$ counts the odd primes that are at least p_i and at most $\frac{n}{3p_i}$. Thus,

$$\Pi_{p_j} \left(\frac{n}{3p_{i+1}} \right)$$

counts the total number of valid choices for p_j when $p = 3$. The function $\Pi_{p_j} \left(\frac{n}{3p_{i+1}} \right)$ gives the total number of odd 3-almost primes with 3 as one of the prime factors. Next, fix $p = 5$, then

$$u = 5p_i q.$$

The constraint is

$$5p_i q \leq n \Rightarrow p_i q \leq \frac{n}{5}.$$

For each fixed p_i , the number of valid choices for q must satisfy

$$q \leq \frac{n}{5p_i}.$$

The function $\Pi_{p_i} \left(\frac{n}{5p_i} \right)$ counts the odd primes that are greater than or equal to p_i and less than or equal to $\frac{n}{5p_i}$. Thus,

$$\Pi_{p_j} \left(\frac{n}{5p_{i+1}} \right)$$

represents the total number of odd 3-almost primes where 5 is one of the prime factors.

Next, repeat the same process for the next odd prime, $p = 7$, then

$$u = 7p_i q.$$

The constraint is:

$$7p_i q \leq n \Rightarrow p_i q \leq \frac{n}{7}.$$

For each fixed p_i , the number of valid choices for q must satisfy:

$$q \leq \frac{n}{7p_i}.$$

The function $\Pi_{p_i} \left(\frac{n}{7p_i} \right)$ counts the odd primes that are at least p_i and at most $\frac{n}{7p_i}$. Thus,

$$\Pi_{p_j} \left(\frac{n}{7p_{i+1}} \right)$$

represents the total number of odd 3-almost primes where 7 is one of the prime factors.

Continue the process for each subsequent odd prime p_j until:

$$\frac{n}{p_j p_{i-j+1}} \leq 3.$$

All identified odd 3-almost primes for each prime p_j are summed. Since primes larger than $\sqrt{\frac{n}{p_j}}$ are excluded, the product of three primes greater than $\sqrt{\frac{n}{p_j}}$ exceeds n . Thus,

define:

$$t = \Pi_{p_j} \left(\left\lfloor \frac{n}{p_j} \right\rfloor^{\frac{1}{2}} \right),$$

where $\Pi_{p_j}(n)$ counts the number of primes from p_j to n . This means t represents the number of odd primes up to $\sqrt{\frac{n}{p_j}}$, which are used to form the odd 3-almost primes.

Hence,

$$\left[\sum_{i=1}^t \Pi_{p_j} \left(\frac{n}{p_j p_{i+j-1}} \right) \right]$$

Since the first prime factor starts from $p_j = 3$, this process continues for each subsequent odd prime p_j . The summation accounts for all valid choices of the second prime factor p_i , ensuring that each term contributes to the total count of 3-almost primes. The process stops when no more valid second prime factors exist. If $t = 0$, then no 3-almost primes can be formed for that choice of p_j . The smallest prime where this happens is p_s , defined as:

$$t = \Pi_{p_j} \left(\left\lfloor \frac{n}{p_j} \right\rfloor^{\frac{1}{2}} \right) = 0.$$

The summation only considers cases where valid second prime factors exist, stopping before p_s so that the count extends only up to p_{s-1} . Hence, the sum of all odd 3-almost primes is

$$\sum_{j=2}^{s-1} \left[\sum_{i=1}^t \Pi_{p_j} \left(\frac{n}{p_j p_{i+j-1}} \right) \right]$$

To avoid over-counting, the number of duplicated counts is subtracted. Each number may be counted more than once when considering different primes. This over-counting occurs for any pair of primes, so the binomial coefficient $\binom{t}{2}$, which accounts for choosing two primes from t primes, is subtracted. This gives:

$$-\binom{t}{2} = -\frac{t(t-1)}{2}$$

Therefore,

$$\Pi^3(n) = \sum_{j=2}^{s-1} \left[\sum_{i=1}^t \prod_{p_j} \left(\frac{n}{p_j p_{i+j-1}} \right) - \frac{t(t-1)}{2} \right]$$

□

Example 3.7. The number of odd 3-almost primes up to n is,

$$\Pi^3(n) = \sum_{j=2}^{s-1} \left[\sum_{i=1}^t \Pi_{p_j} \left(\frac{n}{p_j p_{i+j-1}} \right) - \frac{t(t-1)}{2} \right] \quad (3.6)$$

Let $n = 150$

$$\text{For } j = 2, t = \Pi_3 \left[\left(\frac{150}{3} \right)^{\frac{1}{2}} \right] = \Pi_3(7.07) = 3.$$

$$\text{For } j = 3, t = \Pi_5 \left[\left(\frac{150}{5} \right)^{\frac{1}{2}} \right] = \Pi_5(5.48) = 1.$$

$$\text{For } j = 4, t = \Pi_7 \left[\left(\frac{150}{7} \right)^{\frac{1}{2}} \right] = \Pi_7(4.63) = 0. \text{ Thus, } s = 4$$

Table 3.5: Prime numbers less than or equal to n and corresponding prime counting function $\pi(n)$ values

n	Primes less than or equal to n	$\pi(n)$
16	2, 3, 5, 7, 11, and 13	6
10	2, 3, 5, and 7	4
7	2, 3, 5, and 7	4
6	2, 3, and 5	3

$$\begin{aligned}
\therefore \Pi^3(150) &= \sum_{j=2}^3 \left[\sum_{i=1}^t \Pi_{p_j} \left(\frac{150}{p_j p_{i+j-1}} \right) - \frac{t(t-1)}{2} \right] \\
&= \left[\sum_{i=1}^3 \Pi_{p_2} \left(\frac{150}{p_2 p_{i+2-1}} \right) - \frac{3(3-1)}{2} \right] + \left[\sum_{i=1}^1 \Pi_{p_3} \left(\frac{150}{p_3 p_{i+3-1}} \right) - \frac{1(1-1)}{2} \right] \\
&= \left[\sum_{i=1}^3 \Pi_3 \left(\frac{150}{3 p_{i+1}} \right) - \frac{6}{2} \right] + \left[\sum_{i=1}^1 \Pi_5 \left(\frac{150}{5 p_{i+2}} \right) - \frac{0}{2} \right] \\
&= \left[\sum_{i=1}^3 \Pi_3 \left(\frac{50}{p_{i+1}} \right) - 3 \right] + \left[\sum_{i=1}^1 \Pi_5 \left(\frac{30}{p_{i+2}} \right) - 0 \right] \\
&= \left[\left[\Pi_3 \left(\frac{50}{p_{1+1}} \right) \right] + \left[\Pi_3 \left(\frac{50}{p_{2+1}} \right) \right] + \left[\Pi_3 \left(\frac{50}{p_{3+1}} \right) \right] - 3 \right] + \left[\Pi_5 \left(\frac{30}{p_{1+2}} \right) - 0 \right] \\
&= \left[\left[\Pi_3 \left(\frac{50}{p_2} \right) \right] + \left[\Pi_3 \left(\frac{50}{p_3} \right) \right] + \left[\Pi_3 \left(\frac{50}{p_4} \right) \right] - 3 \right] + \left[\Pi_5 \left(\frac{30}{p_3} \right) - 0 \right] \\
&= \left[\left[\Pi_3 \left(\frac{50}{3} \right) - 3 \right] + \left[\Pi_3 \left(\frac{50}{5} \right) - 3 \right] + \left[\Pi_3 \left(\frac{50}{7} \right) - 3 \right] \right] + \left[\Pi_5 \left(\frac{30}{5} \right) \right] \\
&= [\Pi_3(16.67)] + [\Pi_3(10)] + [\Pi_3(7.14)] - 3 + [\Pi_5(6)] \\
&= [5 + 3 + 3 - 3] + 1 \\
&= (11 - 3) + 1 \\
&= 8 + 1 \\
&= 9
\end{aligned}$$

Therefore, there are 9 of odd 3-almost primes that are less than or equal to 150, and these primes are listed in Table 3.6.

Table 3.6: List of all odd 3-almost primes less than or equal to 150

p	q	r	pq
3	3	3	27
3	3	5	45
3	3	7	63
3	3	11	99
3	3	13	117
3	5	5	75
3	5	7	105
3	7	7	147
5	5	5	125

Chapter 4

Summary and Recommendations

4.1 Summary

1. The formula for finding the number of 2-almost primes less than or equal to n :

$$\pi^{(2)}(n) = \sum_{i=1}^{\pi(n^{\frac{1}{2}})} \left[\pi\left(\frac{n}{p_i}\right) - i + 1 \right]$$

(Theorem 3.1)

2. The formula for finding the number of odd 2-almost primes less than or equal to n :

$$\Pi^{(2)}(n) = \sum_{i=1}^t \Pi\left(\frac{n}{p_{i+1}}\right) - \frac{t(t-1)}{2}$$

(Theorem 3.2)

3. The formula for finding the number of odd 2-almost primes less than or equal to n :

$$\Pi^{(2)}(n) = \pi^{(2)}(n) - \pi\left(\frac{n}{2}\right)$$

(Proposition 3.3)

4. The formula for finding the number of 3-almost primes less than or equal to n :

$$\pi^{(3)}(n) = \sum_{i=1}^{\pi(n^{\frac{1}{3}})} \sum_{j=1}^{\pi\left(\sqrt{\frac{n}{p_i}}\right)} \left[\pi\left(\frac{n}{p_i p_j}\right) - j + 1 \right]$$

(Theorem 3.4)

5. The formula for finding the number of odd 3-almost primes less than or equal to

n :

$$\Pi^3(n) = \sum_{j=2}^{s-1} \left[\sum_{i=1}^t \Pi_{p_j} \left(\frac{n}{p_j p_{i+j-1}} \right) - \frac{t(t-1)}{2} \right]$$

(Theorem 3.5)

4.2 Recommendations

The following are the suggested recommendations for further study:

1. Find more accurate estimations for prime and k -almost prime counts.
2. Determine the relationship between the k -almost primes and primes as n increases infinitely.
3. Find more efficient algorithms to count k -almost primes.

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